

- 6** A particle P of mass 2 kg moving on a horizontal straight line has displacement $x\text{ m}$ from a fixed point O on the line and velocity $v\text{ m s}^{-1}$ at time $t\text{ s}$. The only horizontal force acting on P has magnitude $\frac{1}{10}(2v-1)^2\text{ e}^{-t}\text{ N}$ and acts towards O . When $t = 0$, $x = 1$ and $v = 3$.

(a) Find an expression for v in terms of t .

[5]

[illegible]

(b) Find an expression for x in terms of t .

[4]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.